

```
Exception("Wrong file format at Line = "
+ hexLine);
}
if (hexLine[0] != ':')
//checking for ':'
throw new Exception("No
colon detected");
}
//calculating the length,
offset and type of record
this.length = Convert.
ToInt16(hexLine.Substring(1, 2), 16);
this.offset = Convert.
ToInt32(hexLine.Substring(3, 4), 16);
this.type = Convert.
ToInt16(hexLine.Substring(5, 6), 16);
this.data = new UInt16[this.
length];
/* We now know how long the
record should be */
if (hexLine.Length < (11 +
this.length * 2))
{
throw new
Exception("Wrong hex file format, missing
fields " + hexLine);
}
//do the checksum
checksum = this.length;
checksum += (ushort)((this.
offset >> 8) & 0xff);
checksum += (ushort)(this.
offset & 0xff);
checksum += this.type;
//Parse the data-fields*/
if (this.length != 0)
{
for (recPointer = 0;
recPointer < this.length; recPointer++)
{
this.data[recPointer]
```

```
= (UInt16)Convert.ToUInt16(hexLine.Sub-
string(9 + (recPointer * 2), 2), 16);
checksum +=
(UInt16)this.data[recPointer];
}
//check the correct check-
sum,*/
string sd = hexLine.Sub-
string(9 + (this.length * 2), 2);
UInt16 ui = Convert.
ToInt16(sd, 16);//ui is 2's complement,
checksum
checksum += ui;
StringBuilder alldata = new
StringBuilder();
if ((checksum % 256) != 0)
{
throw new
Exception("Checksum error for HEX record
line" + hexLine);
}
else
{
foreach (UInt16 i in
this.data)
{
alldata.Append(i.
ToString("X2"));
}
return "length-" + this.
length.ToString("X2") + "\t" +
"add-" + this.offset.
ToString("X2") + "\t" +
"Type-" + this.type.
ToString("X2") + "\t" +
"Data-" + alldata.
ToString() + "\t" +
"check-" +
ui.ToString("X2");
}
```

```
}
//Reads and displays all the
contents of the file
if second parameter is true
then detail else listing*/
public string readFile(string
filename, bool List_detail)
{
StringBuilder stb = new
StringBuilder();
try
{
using (StreamReader str
= File.OpenText(filename))
{
while ((hexLine =
str.ReadLine()) != null)
{
if (List_detail
== true)
{
stb.
AppendLine(this.parseRecord(hexLine));//
shows details
}
else
{
stb.
AppendLine(hexLine);//shows the file
}
}
}
}
catch (Exception e)
{
return "Exception " +
e.Message;
}
return stb.ToString();//
shows the file
}
}
```

FORM1.CS

```
/*Below is the Form1.cs file that holds
the program to perform all functions of
the software*/
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Text;
using System.Windows.Forms;
using hexParser;
namespace Hex_File_Parser
{
public partial class Form1 : Form
{
FileParser myFile;
public Form1()
{
InitializeComponent();
myFile = new FileParser();
}
private void btnOpen_
Click(object sender, EventArgs e)
{
if (this.openFileDialog1.
ShowDialog() == DialogResult.OK)
{
this.txtbxFileName.Text
= this.openFileDialog1.FileName;
}
}
private void rdbtList_
Click(object sender, EventArgs e)
{
this.txtbxAddData.Visible
```

```
= true;
if (this.rdbtList.Checked
== true)
{
this.txtbxAddData.Text
= myFile.readFile(this.txtbxFileName.
Text, false);
}
private void rdbtDetail_
Click(object sender, EventArgs e)
{
this.txtbxAddData.Visible
= true;
if (this.rdbtDetail.Checked
== true)
{
this.txtbxAddData.Text
= myFile.readFile(this.txtbxFileName.
Text, true);
}
private void rdbtAddData_
CheckedChanged(object sender, EventArgs
e)
{
if (rdbtAddData.Checked ==
true)
{
string result = myFile.
readFile(this.txtbxFileName.Text);
if (result == "Suc-
cess")
{
Int32 start = my-
```

```
File.
Int32 end = myFile.
Fend;
Int32 range = end
- start;
StringBuilder
stb=new StringBuilder();
for (int j = 0, i =
start; j <= (range >> 1); j++)
{
stb.
AppendLine("add -> " + (i >>
1).ToString("X2") + " " +
"high -> " +
myFile.filedata[i + 1].ToString("X2") +
" " +
"low -> " + my-
File.filedata[i].ToString("X2"));
i = i + 2;
}
this.txtbxAddData.
Text=stb.ToString();
}
else
{
this.txtbxAddData.
Text = result;
}
}
private void butClose_
Click(object sender, EventArgs e)
{
this.Dispose(true);
}
}
```